0014 — Trend-Filtered Short-Term Mean Reversion on a Colored Stochastic — 2026-07-08

Series: MC Setup Research Notes · Note 0014 **Confidence:** Medium-High **as a positive result** — 17 years of ES daily bars (2009-04 → 2026-07, 4,442 sessions), one economically-motivated parameter choice, walk-forward validated (12-month in-sample / 4-month out-of-sample), robust in 16 of 17 calendar years. Exploratory (not pre-registered); single instrument, single timeframe. The headline dollar figures below are shown on **micro (MES) and unlevered** sizing, not a full ES contract, so they reflect what a real account can carry.

TL;DR: Buy the ES on the daily close when a short stochastic is deeply oversold and price is above its 100-day average; sell when price closes back above its 5-day average. Enter market-on-close in the last minutes (that beats the next-day open by ~11%). Over 17 years this fires ~11×/year, wins **80%**, runs a **profit factor of 4.45**, and is positive in **16 of 17 years**. On **1-lot MES** it makes ~**+$19.4k** with a worst-ever drawdown of **−$808** on a ~**$10k** account; expressed unlevered (ETF shares, any account size) it compounds **×2.58** with a max drawdown of **−4.8%**. A **regime-gated short** (mirror signal, only in confirmed downtrends) turns the system all-weather and paid **+$87k-equivalent in 2022**. The oscillator is the trader's own indicator (an 8-period colored stochastic); a textbook 14/3/3 version does not work — the specific settings matter.

1. The setup (so this note stands alone)

Instrument / data. ES continuous futures, **RTH daily bars** (settlement stamp 15:15 CT), 2009-04-13 → 2026-07-06. Indicator values are exported bar-by-bar from NinjaTrader by MyStochasticExporter (self-contained; reproduces MyStochasticsColorwithSignal math): **%K = 100 × Σ(Close−LowN) / Σ(HighN−LowN)** over **N=8** bars, smoothing 1 (so %D=%K). The "colored" oversold zone is %K in its lower band.

The system (final).

- **LONG entry:** %K(8) < 15 **AND** Close > SMA100. Fill at the **signal-day close** (market-on-close, last minutes).
- **Exit:** first day that Close > SMA5; hard time stop 15 bars.
- **Regime SHORT (optional bear hedge):** %K(8) > 85 **AND** SMA50 < SMA200 **AND** Close < SMA50; exit first Close < SMA5.

What it is, in pros' language. A trend-filtered short-term mean-reversion (STMR) swing system — the "Connors school": buy short-term oversold inside a longer uptrend. Colloquially, buy-the-dip / fade-the-rip.

2. The questions

1. Does the trader's colored stochastic carry a real, standalone mean-reversion edge on the daily? 2. What is the right oscillator setting — and does optimizing further help or overfit? 3. Which trend filter (length; SMA vs EMA) controls risk, and why did 2011/2018/2022 hurt? 4. Does entering at the close (MOC) beat the next-day open? 5. Does adding the short side help in a prolonged bear — and how must it be gated? 6. What does this look like on an account you can actually fund (MES / unlevered), not a full ES contract?

3. How we tested it

- **Backtest:** event-ordered on daily OHLC, entry next-open **and** signal-close variants, ~$4 round-turn (ES) / ~$1.50 (MES). 1 contract per signal; concurrency tracked for margin.
- **Parameter sweep:** K in {5,8,10,14,21}, smoothing in {1,3}, oversold in {10,15,20,25}; ranked by **per-year robustness**, not headline PF.
- **Walk-forward:** 12-month IS optimize → 4-month OOS, rolled 2010→2026; compared to the fixed setting.
- **Trend filter:** SMA100/150/200/250, EMA200, plus regime gates (SMA50>SMA200, C>SMA50, slope).
- **Sizing:** 1 ES, 1 MES, and unlevered %-compounded (ETF-share equivalent).

4. Results

4.1 The edge is real — and it's the trader's own stochastic

A self-rolled generic 14/3/3 daily stochastic produced garbage (n≈15, negative). The trader's **8-period oversold zone** is a genuine edge, standalone PF ≈ 2.1 and, as a confirm on RSI-2, lifts expectancy **+24%** without adding tail risk. The specific short, fast setting is doing the work.

4.2 One sensible knob; more optimization overfits

Tightening the oversold line **20** → **15** on the default K=8 is a single, economically-sensible improvement (demand a deeper oversold): PF 2.08 → **2.83**, positive **15/17** years. Sweeping K and OS jointly finds flashier combos (PF 3.7) that fire only ~100 times and miss a year — classic curve-fitting.

4.3 Walk-forward says: lock the parameters

Approach	OOS PF	OOS total (ES)	OOS max DD	Return/DD
Re-optimize K & OS every 4 months	2.11	+$165k	−$47.2k	3.5
Fixed setting (no re-opt)	**2.71**	**+$191k**	**−$16.8k**	**11.3**
Re-optimize trend length	3.56	+$151k	−$13.3k	11.3
Fixed SMA100	**3.77**	**+$159k**	**−$7.9k**	**20.2**

Re-optimization **degrades** out-of-sample every way we cut it. The rolling optimizer's picks are unstable (noise-chasing on ~15 sparse signals per window). **Fix the parameters and stop tuning** — the honesty check that most retail systems fail.

4.4 Trend filter, and why 2011/2018/2022 hurt

Trend gate (entry %K8<15)	n	PF	max DD (ES)	2011	2022
C > SMA200	253	2.85	−$16.8k	−$6.3k	−$12.1k
C > SMA100	184	**4.02**	**−$7.9k**	**+$5.2k**	**+$3.5k**
C > SMA200 & C > SMA50	100	7.75	−$5.1k	+$2.1k	+$2.3k

Late-2011 (US downgrade + Euro crisis), Q4-2018 and 2022 were **whipsaws around the 200-day** — brief pops above the line tripped "buy the dip," then price plunged. A single slow average can't tell an uptrend from a bear-market bounce. A **nearer trend line (SMA100)** — or a second gate (C>SMA50) — refuses those dips and turns all three crisis years positive. EMA ≈ SMA. Loosening the exit to SMA10 was catastrophic (drawdown −$107k): let losers run and a mean-reversion system dies.

4.5 Enter at the close, not the open

Entry	PF	exp/trade (ES)	max DD	Return/DD
next-day OPEN	4.02	+$962	−$7.9k	22.4
signal CLOSE / MOC	**4.45**	**+$1,064**	−$8.1k	**24.3**

The market-on-close entry captures the **overnight bounce** that tends to follow an oversold settlement, worth **~+11%** expectancy for free. (Mild caveat: it uses the forming settlement to trigger — execute in the final minutes.) SPX options also trade to 15:15 CT, so the same last-minutes window works for an options expression.

4.6 The bear hedge — right instinct, must be gated

A naive symmetric short (%K8>85 & C<SMA100) is a **loser that blew up −$85k in the 2026 rally** (shorting into a powerful uptrend). Gated to a **confirmed downtrend** (SMA50<SMA200 & C<SMA50) it becomes a real hedge: 2022 **+$34–87k**, 2026 ≈ flat. Combined **long+short** doubles gross to **+$228k (ES)** and is positive **16/17** years — at lower Return/DD than long-only. Trade-off: smoother (long-only) vs all-weather (add the gated short).

4.7 What it costs to trade it (the number that matters)

LONG core, MOC close, 184 trades / 17 years, **1 contract at a time**:

Vehicle	exp/trade	17-yr total	max DD	worst trade	min account	scalable
1 ES ($50/pt)	+$1,064	+$196k	−$8.1k	−$8.1k	~$60k	coarse
1 MES ($5/pt)	**+$105**	**+$19.4k**	**−$808**	−$808	**~$10.4k**	per lot
Unlevered (ETF shares)	**+0.52%**	**×2.58**	**−4.8%**	−4.1%	**any**	fractional

Peak concurrency is **5** positions → peak MES margin ~$6,600; a **~$10k account trades 1-lot MES** with a heat buffer, adding lots as it grows. Unlevered, it compounds **×2.58** at **~6%/yr on deployed capital** with a **−4.8%** max drawdown — and because it is **in-market only ~9% of days**, the idle 91% can earn T-bills (~4–5%), materially lifting the blended return. This is a **"cash + occasional dip-buy"** profile, not full-time market exposure.

5. Why it works

US equity indices are structurally upward-drifting and, on the daily, short-horizon **overreaction reverts**: a sharp oversold close inside an uptrend is a liquidity/emotion flush that the drift repairs within days. The trend filter keeps you on the side of the drift; the fast oversold oscillator times the flush; the close entry harvests the overnight repair; the SMA5 exit takes the reversion and leaves. The system's weakness is the mirror of its strength — in a sustained bear the drift is gone, so the trend gates flatten activity, and the residual danger is the choppy topping/whipsaw phase (mitigated, not eliminated, by the SMA50/100 gates).

6. What could kill it / limitations

- **Crash tail:** a long-only dip-buyer cannot be fully immunized; worst single-trade heat ≈ **−$18k per ES contract** (March-2020). Defenses: **vol-scaled sizing (ATR)**, the trend gates, small size, or the defined-risk options version.
- **Single instrument / timeframe.** Intraday mean-reversion on ES tested negative earlier; weekly is too sparse to trade alone. A weekly-uptrend confirm marginally lifts PF (4.45→4.98) but isn't required.
- **Edge decay:** these tools are now widely available; the durable advantage is process, sizing, execution, and diversification across instruments (NQ/RTY/ZN/GC), not the backtest.
- **Options numbers are unverified** — a bull-put-spread expression is attractive (sell rich IV into the oversold spike) but our prior PnL was a Black-Scholes estimate; validate on real chains (OptionsDX/DoltHub) before trusting.

7. How to trade it (spec)

1. Daily ES/MES. Compute %K(8), SMA5, SMA50, SMA100, SMA200 on RTH closes. 2. **Long** when %K8<15 & Close>SMA100; enter **MOC** in the last minutes. Exit next day Close>SMA5 (time-stop 15 days). 3. **Optional short** when %K8>85 & SMA50<SMA200 & Close<SMA50; exit Close<SMA5. 4. Size on **MES** to your account (1 lot ≈ $10k, add lots as equity grows) or trade **unlevered ETF shares** with idle cash in T-bills. 5. **Do not re-optimize** the parameters — walk-forward shows fixed beats fitted.

Reproduce: python scripts/mr_stoch_system.py · data data/ES_stoch_daily.csv · exporter nt8/indicators/MyStochasticExporter.cs · figures docs/living/mr_stoch_system.png, mr_stoch_realistic.png, mr_stoch_wfa.png, mr_bothsides.png.